

## (a) CQG-Single

One-step continuous query generation

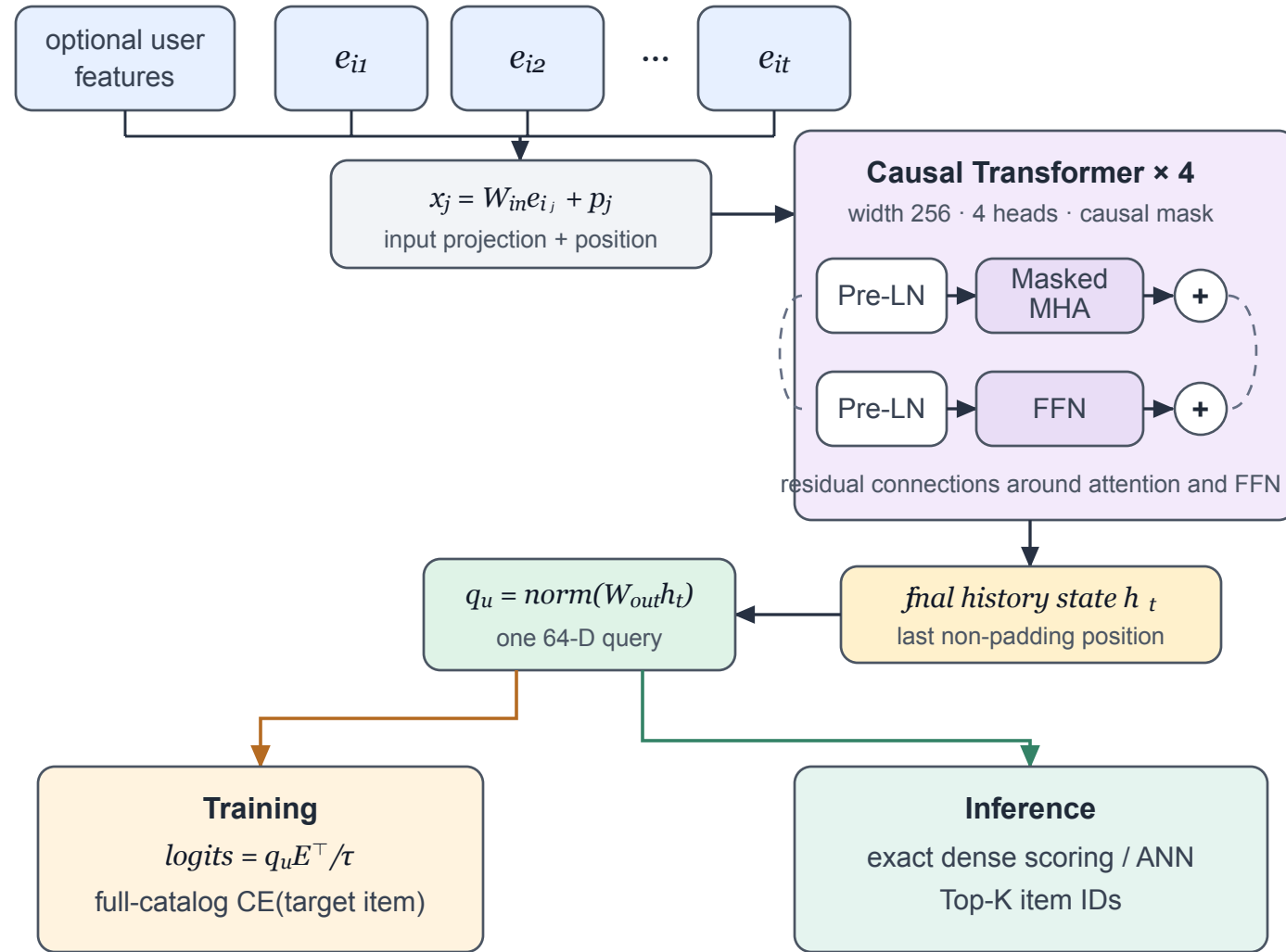
one forward pass

### CQG-Single output interface

one 64-D query · no SID tokens · no serial decoding · no beam expansion

#### User context

history-item embeddings from frozen E



## (b) CQG-AR

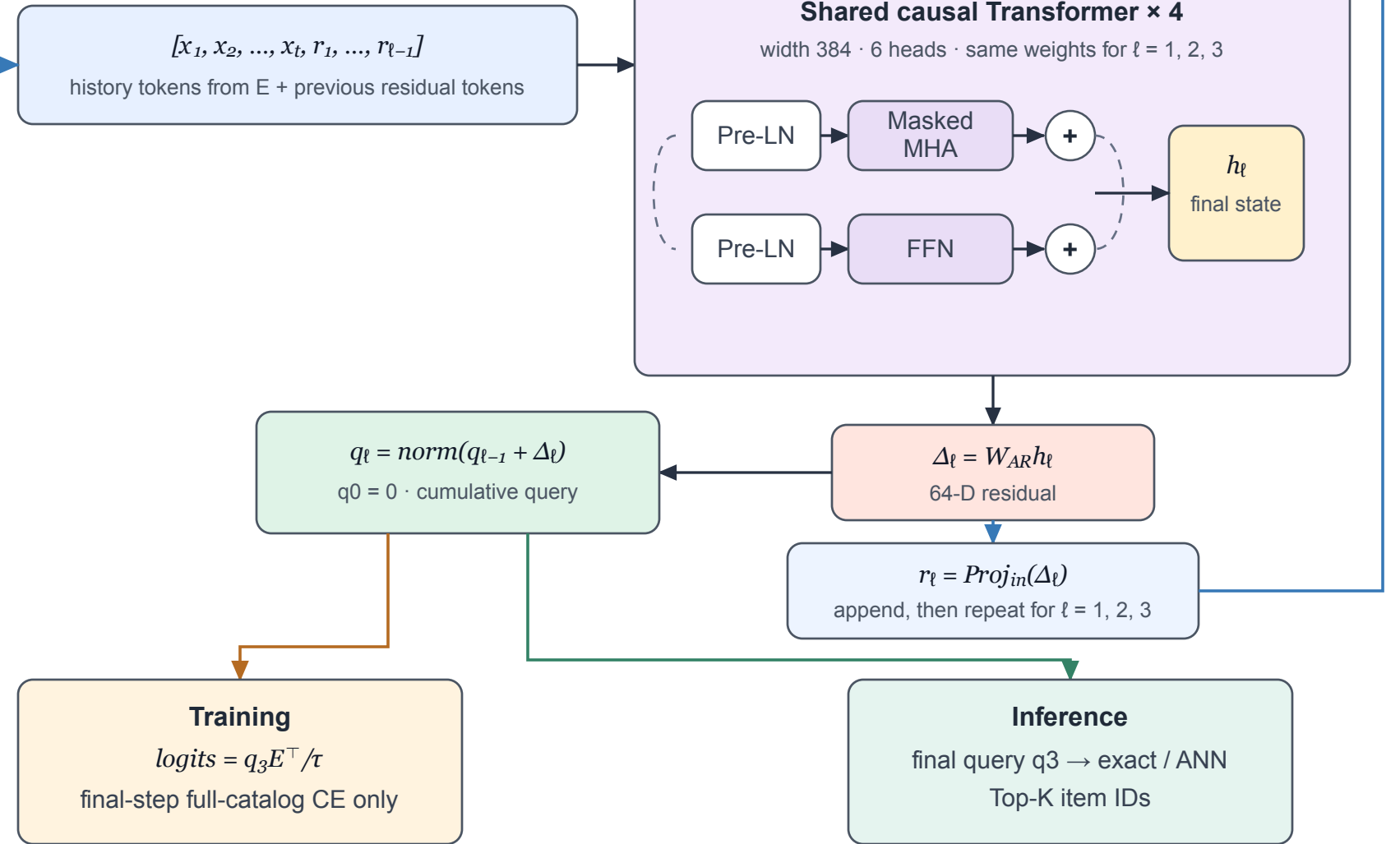
Autoregressive continuous residual refinement

L = 3 decoding steps

### CQG-AR output interface

continuous residual sequence → final 64-D query · no SID tokens · no code lookup · no beam expansion

#### Autoregressive input at step $\ell$



→ forward computation → AR feedback only → training → retrieval

## Shared frozen collaborative item space

$$E = [e_1, \dots, e_N]^\top \in \mathbb{R}^{N \times 64}$$

① history-item lookup

② full-catalog classifier weights

③ exact / ANN retrieval index